

ROAD MAP TO FUTURE FACTORIES (FY26-28)

• India will not only miss a historic window of opportunity if it fails to adopt frontier technologies but also encounter a potential loss of additional manufacturing GDP to an extent of \$270 billion by 2035 and \$1 trillion by 2047. Here are a few focus areas



Digital Twins

Virtual, data-driven replicas of physical systems which should become the digital backbone of Indian manufacturing



Robotics & Automation

Transition from fixed automation to intelligent collaborative systems across shop floors



AI Integration

AI will be the core layer of manufacturing, embedded across production lines, logistics networks, design systems, and product interfaces



Cluster Development

Regional hubs (semiconductor clusters, electronics parks) to pool resources and talent, and enable ease of coordination and access



Plug & play tech enabled industrial parks

These will enable MSMEs to access and experiment with AI, ML, robotics, etc



Policy Support

PLI schemes, SEZ relaxations and India Semiconductor Mission incentives

NOTE ABOVE LIST IS NOT EXHAUSTIVE, BUT INDICATIVE
SOURCE NITI AAYOG, REIMAGINING MANUFACTURING REPORT 2025

fore a breakdown disrupts production. Automated visual inspection systems use computer vision to detect defects in components faster and more accurately than human workers.

Vinod Sharma, chairman of the Confederation of Indian Industry's (CII) National Committee on Electronics Manufacturing, says, "Such technologies are reshaping shop floors. AI-powered sensors can forecast machine failures before they occur, reducing downtime dramatically. Computer vision systems can identify micro-defects on circuit boards that would be almost impossible for the human eye to detect."

A HYBRID MODEL

Industry experts like Gujral say the country's manufacturing ecosystem will likely have hybrid factories rather than fully automated dark facilities due to availability of skilled labour and the high cost of building and running dark factories.

India's likely path is hybrid manufacturing, combining low-cost labour with targeted automation. "Labour costs are relatively low and availability of manpower is high. But areas where fatigue affects quality or where human judgement becomes inconsistent will increasingly be automated," says Gujral.

One example is visual inspection. On electronics assembly lines, workers often inspect components for defects. But after an hour of staring at tiny circuit boards, the human eye begins to tire. Automated optical inspection systems can perform this task far more consistently. As a result, such processes are among the first to be automated.

Electronic manufacturing companies have started using robots. However, the penetration gap compared to global levels is still very wide, says Gokul of CynLr.

According to the International Federation of Robotics, India's robot density is significantly lower than that of major manufacturing economies such as South Korea, China, Japan, Germany, and the US. Robot density is the number of robots per 10,000 human employees. India has a robot density of 30. South Korea is at 1,220 and China at 166, while the global average is 132, according to the report.

Sharma, who is also managing director of